

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester III Examination, 2022-23

Paper No. BCH-216: Basic Statistics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. All questions carry equal marks (14 marks each).***Question 1.**

“The successful businessman is the one whose estimates most closely approaches accuracy.”
 Explain and discuss the limitations of Statistics. [14]

OR

Draw a cumulative frequency curve of the distribution given below and read the values of the median and two quartiles:

oprule Marks	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	4	11	15	24	10	6
ottomrule						

[14]**Question 2.**

Explain the relationship among arithmetic mean, geometric mean and harmonic mean. [14]

OR

(a) Gains and losses (in Rs.) of 50 speculators are given below:

- 4 speculators gain between Rs. 300 and Rs. 400
- 6 speculators gain between Rs. 200 and Rs. 300
- 8 speculators gain between Rs. 100 and Rs. 200
- 12 speculators gain between Rs. 0 and Rs. 100
- 10 speculators lose between Rs. 0 and Rs. 100
- 7 speculators lose between Rs. 100 and Rs. 200
- 3 speculators lose between Rs. 200 and Rs. 300

Find out the average gain of gainers, average loss of losers and average net gain or loss of the whole group. [7]

(b) Compute the mean and mode marks of all the students of 50 colleges in a District given below:

oprule Marks obtained	No. of colleges	Average number of students in a college
35 and above	7	200
30–35	10	250
25–30	15	300
20–25	9	200
15–20	5	150
Less than 15	4	100
ottomrule		

[7]

Question 3.

The following data are related with the mileage of two types of tyres:

oprule Life (in km)	No. of tyres: Type A	No. of tyres: Type B
20000–22000	230	200
22000–24000	270	325
24000–26000	450	470
26000–28000	375	300
28000–30000	125	155
ottomrule		

(a) Which of the two types give a higher average life?

[7]

(b) If the price of these two types are same, which one will you prefer and why?

[7]

OR

Calculate Karl Pearson's coefficient of skewness from the following data:

oprule Marks	No. of students
Less than 20	8
Less than 30	38
Less than 40	53
40–50	2
50–60	8
60–70	30
More than 70	17
More than 80	4
ottomrule	

[14]

Question 4.

For 50 paired observations of X and Y , the following data were obtained:

$$arX = 10, \quad \sigma_x = 3, \quad arY = 6, \quad \sigma_y = 2, \quad r_{xy} = 0.3$$

But on subsequent verification it was found that in a pair one value of $X (= 10)$ and one value of $Y (= 6)$ were inaccurate and hence needed out. With the remaining 49 pairs of values, how is the original value of correlation coefficient affected?

[14]

OR

Distinguish clearly among simple, partial and multiple correlation and point out their usefulness in statistical analysis. [14]

Question 5.

Explain regression coefficients. How are the regression coefficients of two regression lines related with the correlation coefficient? [14]

OR

From the following data:

Age of husband (Year)	Age of wife (Year)
18	17
19	17
20	18
21	18
22	18
23	19
24	19
25	20
26	21
27	21

Find

find:

- (a) the regression equation of age of husband on the age of wife; [4]
- (b) the regression equation of age of wife on the age of husband; [4]
- (c) total variation in the age of husband; [3]
- (d) the magnitude of variation in age of the husband explained by the regression equation. [3]